

CREATIVE INSTITUTE DATA SCIENCE

Normal Distribution and Z Score: Math and statistics for data science

```
import pandas as pd
import seaborn as sn
height=[6.2,5.7,4.6,6.4,5.9,4.0,5.1,5.2,5.9]
sn.histplot(height,kde=True)
```

How Normal Distribution used in data analysis

Data Cleaning: Outlier removal

We are going to use heights dataset from kaggle.com. Dataset has heights and weights

<https://www.kaggle.com/mustafaali96/weight-height>

outlier that means error or we can say outlier is data set which is away from group

```
import pandas as pd
import seaborn as sn
df=pd.read_csv('heights.csv')
df
df.head()
```

Outlier detection and removal using Standard Deviation

```
df.Height.describe()
```

```
sn.histplot(df.Height, kde=True)
```

```
mean = df.Height.mean()
mean
```

```
std_deviation = df.Height.std()
std_deviation
```

for remove outlier normally use 3 standard deviation

rules for 3 std dev.

```
mean- 3*std_deviation
mean+ 3*std_deviation
```

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for this example we can say outlier is if any of heights is lower than mean minus 3 std deviation or heights is higher than plus 3 std deviation

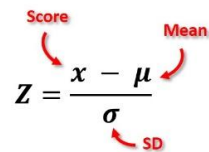
```
df[(df.Height < 54.82) | (df.Height > 77.91)]
```

if you want cleaning dataset without outlier

```
df_no_outlier = df[(df.Height<77.91) & (df.Height>54.82)]  
df_no_outlier.shape
```

Z Score

how many standard deviation away a datapoint is from mean

$$Z = \frac{x - \mu}{\sigma}$$


X=individual data point

Add new column name zscore in data frame

```
df['zscore'] = ( df.Height - df.Height.mean() ) / df.Height.std()  
df.head(5)
```

```
df.Height.mean()
```

```
df.Height.std()
```

```
(73.84-66.37)/3.84
```

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If any dataset of zscore > 3 is outlier or else dataset zscore is less 3 is outlier

```
df[df['zscore']>3]
```

```
df[df['zscore']<-3]
```

Exercise

(Normal Distribution and Z Score for Outlier Removal)

You are given bhp.csv which contains property prices in the city of banglore, India.

You need to examine price_per_sqft column and do following,

1. Remove outliers using percentile technique first. Use [0.001, 0.999] for lower and upper bound percentiles
2. After removing outliers in step 1, you get a new dataframe.
3. On step(2) dataframe, use 4 standard deviation to remove outliers
4. Plot histogram for new dataframe that is generated after step (3). Also plot bell curve on same histogram
5. On step(2) dataframe, use zscore of 4 to remove outliers. This is quite similar to step (3) and you will get exact same result